

Topic: Principal Component Analysis (PCA) and Applications

Required Content

- Study the Principal Component Analysis (PCA) method based on eigenvalues and eigenvectors of the covariance matrix.
- Apply PCA to data dimensionality reduction and visualization.

Description

- Construct the covariance matrix.
- Compute eigenvalues and eigenvectors.
- Select principal components corresponding to the largest variances.
- Perform data dimensionality reduction.
- Apply PCA to real-world datasets.

Expected Results

- Implement the PCA algorithm from scratch.
- Visualize data before and after dimensionality reduction.
- Analyze and interpret the meaning of the principal components.

Notes

- Do not directly use built-in PCA libraries.